

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (MCA)

Semester-I University Examination 2009-10

CA701 Fundamentals of Programming

Date: 24.12.09, Thursday Time: 01 p.m to 04 p.m

Maximum Marks: 70

Instructions:

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION-1

Q1(a) Differentiate following terms

06

- (i) Procedure language Vs Non-procedure language.
- (ii) Continue Vs Break.
- (iii) Entry Control loop Vs Exit Control loop.

(b) State whether the following statements are true or false. Give 04
Justification for each statement.

- (i) Modulus operator can be used for integer and real type of data.
- (ii) Operand in increment operator may be variable or constant.
- (iii) Associativity of ternary operator is from Left to Right.
- (iv) In switch statement, all cases automatically break after execution of the corresponding statement list for case label.

(c) Give the output of following:

05

(i)

```
void main()
{
    printf( "\n gujarat");
    printf( "\b state");
    printf( "\r india");
}
```

(ii)

```
void main()
{
    char s[] = " CHARUSAT";
    printf( "%12s\n",s);
    printf( "% .2s\n",s);
}
```

(iii)

```
void main()
{
    int i=9,j=3;
    float num= 45.56;
    printf( ".*.f", i,j,num);
}
```

(iv)

```
void main()
{
    int mat [] = { 1,2,3,4,5,6,7};
    printf( "%d", mat[ mat[1] ] );
}
```

(v)

```
void main()
{
    int num=9;
    while ( num )
    {
        printf( "%d",num);
        num - - ;
    }
}
```

Q2(a) What is meant by a flow-chart? Explain it with proper example.

05

(b) Explain while loop.

04

(c) Define: Compiler.

01

OR

[P.T.O]

- Q2 (a) Explain switch statement in detail. 05
 (b) Discuss the features of structured programming. 04
 (c) Differentiate variable and constant. 01
 Q3 (a) Write a C program to determine whether the given number is prime or not. 05
 (b) Explain basic data types of C. 04
 (c) What is the purpose of sizeof() operator? 01

OR

- Q3 (a) Write a C program to calculate average of first N even numbers. 05
 (b) Explain arithmetic operators of C language. 04
 (c) Give difference of getch() and getche(). 01

SECTION-2

- Q4 (a) Write a C program to determine position of given element, if element is present in list, other wise display proper message. (Use binary search) 08
 (b) Explain any two string manipulation functions. 04
 (c) Define following terms : 03
 (i) Data Structure (ii) Priority Queue (iii) Double ended Queue
 Q5 (a) What is mean by stack? Explain algorithm of all operations of stack. 08
 (b) What is the purpose of gets () function. 01
 (c) What is the purpose of atoi() function. 01

OR

- Q5 (a) List out applications of stack. Explain algorithm to convert infix notations to prefix and postfix with example. 08
 (b) Differentiate scanf and sscanf functions. 01
 (c) List out any five functions available in ctype.h header file. 01
 Q6 (a) Write a short note on : Bubble Sort 05
 (b) What is mean by array? Can array indexes be negative? Demonstrate the storage of one-dimensional array in memory with help of diagram. 04
 (c) What is limitation of queue compare to circular queue? 01

OR

- Q6 (a) Write a C program to merge two strings without any string manipulation function. 05
 (b) Write a short note on: Selection Sort. 04
 (c) Define: Queue. 01